

Data Mining

Definition of Data Mining

Data mining is the process of extracting hidden, previously unknown, and useful information from large amounts of data.

It is a **non-trivial process**, which means it involves advanced techniques and intelligent analysis to discover meaningful patterns.

In simple words, data mining helps to find valuable knowledge from huge databases.

Data mining is the search for:

- Relationships
 - Patterns
 - Trends
- that exist in large databases but are hidden among large amounts of data.

For example:

Finding the relationship between:

- Patient symptoms and diseases
- Customer behaviour and product sales

These relationships provide important knowledge that helps in decision making.

Techniques Used in Data Mining

Data mining includes many technical approaches such as:

1. **Clustering** – Grouping similar data together.
2. **Classification** – Assigning data into predefined categories.
3. **Data summarization** – Providing a summary of large data.
4. **Dependency analysis** – Finding relationships among data items.
5. **Change and trend analysis** – Studying how data changes over time.
6. **Anomaly detection** – Identifying unusual or abnormal data.

These techniques help organizations analyze data effectively.

Purpose of Data Mining

Data mining is used to identify important information or decision-making knowledge from data and apply it in areas such as:

- Decision support
- Prediction
- Forecasting
- Estimation

Thus, it helps organizations to improve planning and performance.

Why is Data Mining Important?

Data mining plays an important role in modern organizations. It is a key part of analytics and business intelligence systems.

Supports Decision Making -It helps managers make better and faster decisions.

Business Intelligence -The extracted information is used in business intelligence and advanced analytics.

Predict Future Trends -Organizations can predict future events based on historical data.

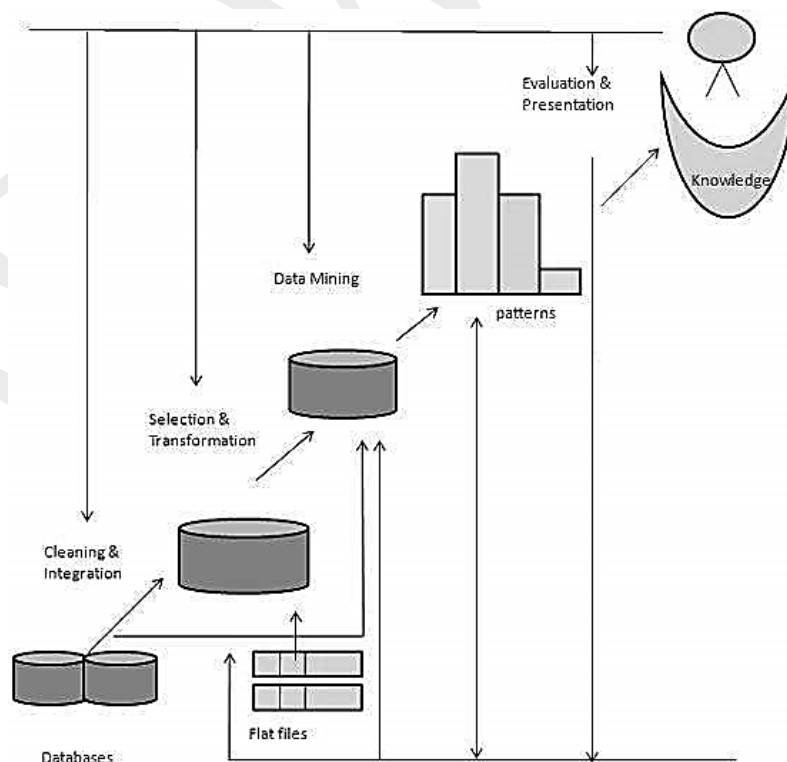
Real-Time Analysis - It supports real-time analytics by analyzing streaming data.

Improves Efficiency and Profit - Helps in identifying customer needs and improving business strategies.

Risk Management - Detects fraud, security threats, and abnormal activities.

Data mining is an important process that extracts useful knowledge from large datasets. It helps organizations in decision making, prediction, and improving overall performance. Therefore, it is widely used in business, healthcare, banking, and many other fields.

KDD – Knowledge Discovery in Databases



Knowledge Discovery in Databases (KDD) is a field of computer science that includes tools and techniques used to extract useful, valid, and previously unknown information (knowledge) from

large collections of data. It is a complete process that helps humans discover meaningful patterns and relationships from digital data. **Data mining is only one step in the KDD process.**

Steps in KDD Process

1. Data Cleaning

In this step, noise and inconsistent data are removed from the database.

This includes:

- Handling missing values
- Removing duplicate data
- Correcting errors
- Eliminating irrelevant information

Example:

Incorrect or missing age values in a customer database are corrected.

This step improves the quality and accuracy of data.

2. Data Integration

In this step, data from multiple sources are combined into a single database.

Sources may include:

- Different databases
- Files
- Data warehouses
- Web data

Example:

Combining customer data from different branches of a company.

This step ensures consistency and completeness.

3. Data Selection

In this step, only relevant data required for analysis is selected.

This reduces:

- Processing time
- Complexity
- Storage requirements

Example:

Selecting only sales data for a specific year.

4. Data Transformation

In this step, data is converted into a suitable format for mining.

This includes:

- Normalization
- Aggregation
- Summarization
- Data consolidation

Example:

Converting marks from different scales into a common format.

This step makes the data suitable for analysis.

5. Data Mining

In this step, intelligent techniques are applied to extract patterns and knowledge.

Some methods used:

- Classification
- Clustering
- Association
- Prediction

This is the core step of the KDD process.

6. Pattern Evaluation

In this step, the discovered patterns are evaluated to find:

- Valid
- Useful
- Interesting patterns.

Irrelevant or unnecessary patterns are removed.

7. Knowledge Presentation

In this final step, knowledge is presented in a user-friendly way.

This includes:

- Graphs
- Charts
- Tables
- Reports
- Visualization tools

This helps users understand and make decisions.

Thus, KDD is a systematic and iterative process used to extract meaningful knowledge from large databases. It includes several steps such as data cleaning, integration, selection, transformation, mining, evaluation, and presentation. Data mining is only one part of this complete process.

Difference between DBMS and Data Mining

Sl. No	Basis	DBMS (Database Management System)	Data Mining
1	Meaning	DBMS is used to store, manage and retrieve data from databases.	Data mining is the process of extracting hidden, previously unknown and useful information from large databases.
2	Nature of Data	Deals with explicitly stored data.	Deals with implicit and hidden patterns in data.
3	Main Objective	Focuses on data storage and retrieval.	Focuses on knowledge discovery and pattern identification.
4	Type of Processing	Query-based processing using SQL.	Analytical and exploratory processing.
5	Information Extraction	Retrieves only the stored data.	Extracts relationships and global patterns from data.
6	Techniques Used	Queries, indexing, and transaction processing.	Statistical methods, machine learning and pattern recognition techniques.
7	Output	Produces reports, tables, and records.	Produces patterns, trends, correlations and predictions.
8	Decision Support	Does not directly support decision making.	Strongly supports decision making, prediction and forecasting.
9	Learning Ability	Does not learn from past data.	Self-learns from previous data and improves accuracy.
10	Example	Retrieving student marks above 80%.	Predicting which students are likely to fail.

Difference between KDD and Data Mining

Knowledge Discovery in Databases (KDD) and Data Mining are closely related terms. Many people use them interchangeably, but they are not the same. Data mining is only a part of the complete KDD process.

Data Mining is a Step in KDD

Data mining is only one of the many steps involved in the KDD process. It focuses mainly on applying algorithms and techniques to extract patterns from data.

On the other hand, KDD is a complete process that includes:

- Data cleaning
- Data integration
- Data selection
- Data transformation
- Data mining
- Pattern evaluation
- Knowledge presentation

Thus, KDD is broader, while data mining is a specific step within it.

KDD is Iterative and Interactive

The KDD process is highly iterative and interactive. This means:

- The steps are repeated multiple times.
- Users interact with the system to improve the results.
- The process continues until useful knowledge is discovered.

For example:

If the results are not satisfactory, the user may:

- Clean the data again
- Select different data
- Apply new mining techniques

This improves the quality of knowledge.

Conditions for Knowledge in KDD

The patterns discovered in data mining must satisfy certain conditions to be considered useful knowledge. These conditions are:

Validity : The discovered patterns must be accurate and correct. They should work well on new or unseen data.

Understandability: The patterns should be simple and easy to understand by users. If the results are complex, they may not be useful.

Utility: The knowledge should be useful in solving real-world problems.
It should help in decision making.

Novelty: The discovered information should be new and not already known.

Interestingness: The patterns should be meaningful and interesting to users.
They should provide valuable insights.

In conclusion, data mining is an important step in the overall KDD process. While data mining extracts patterns from data, KDD ensures that these patterns become useful knowledge by following a systematic and iterative process. The discovered knowledge must be valid, understandable, useful, novel, and interesting.

Elements and Uses of Data Mining

Data mining is a systematic process that involves several important steps to extract useful knowledge from large databases. It consists of the following five major elements:

1. Extract, Transform and Load (ETL) Data into the Data Warehouse

The first step in data mining is to collect data from different sources such as:

- Databases
- Files
- Websites
- Transaction systems

This data is then:

- **Extracted** from various sources
- **Transformed** into a suitable format
- **Loaded** into a data warehouse

This process is called **ETL (Extract, Transform, Load)**.
It ensures that the data is clean, consistent, and ready for analysis.

2. Store and Manage Data in a Multidimensional Database

After loading, the data is stored and managed in a **multidimensional database system**.

This helps in:

- Organizing data in the form of dimensions such as time, location, and product
- Supporting fast data analysis
- Improving performance of queries

Example:

Sales data can be stored based on:

- Time
- Product
- Region

This structure helps in better decision making.

3. Provide Data Access to Analysts and IT Professionals

The stored data must be easily accessible to:

- Business analysts
- Managers
- IT professionals

They use the data to:

- Understand trends
- Generate reports
- Support decision making

Tools such as:

- OLAP (Online Analytical Processing)
- Dashboards
- Reporting tools
are used to access the data.

4. Analyze the Data using Application Software

In this step, different **data mining algorithms and software** are used to analyze the data.

These include techniques such as:

- Classification
- Clustering
- Association
- Prediction

The purpose is to:

- Identify hidden patterns
- Discover relationships
- Predict future trends

This helps organizations make better business strategies.

5. Present the Data in a Useful Format

The final step is to present the analyzed data in an understandable and meaningful form.

The results are shown using:

- Graphs
- Charts
- Tables
- Reports
- Dashboards

This makes it easier for users to:

- Interpret the results
- Take proper decisions
- Improve performance

Data Mining Techniques

Data mining techniques are the methods used to extract useful information and hidden patterns from large amounts of data. These techniques help organizations in decision making, prediction, and understanding customer behaviour.

The important data mining techniques are:

1. Data Cleaning and Preparation

Data cleaning and preparation is an important step in the data mining process. Raw data is usually incomplete, noisy, or inconsistent. So, it must be cleaned and formatted before analysis.

This process includes:

- Removing missing or incorrect values
- Eliminating duplicate data
- Formatting the data into a suitable structure
- Transforming data for analysis

It also includes activities such as:

- Data modelling
- Data transformation
- Data integration
- Data migration
- ETL (Extract, Transform, Load)
- Data aggregation

Therefore, data cleaning and preparation help in understanding the basic features of data and improve the accuracy of data mining results.

2. Tracking Patterns

Tracking patterns is a basic and important data mining technique. It involves identifying and monitoring trends or patterns in data.

This helps organizations:

- Understand customer behaviour
- Identify sales trends
- Make better business decisions

For example:

If a company finds that a particular product is selling more among a specific group of customers, it can:

- Increase the stock of that product
- Develop similar products
- Target advertisements to that group

Thus, tracking patterns helps in improving business performance.

3. Association

Association is a data mining technique used to find relationships between data items.

It shows that certain events or data are connected to other events.

For example:

In a supermarket, customers who buy bread may also buy butter.

This technique is mainly used in:

- Market basket analysis
- Recommendation systems
- E-commerce websites

Therefore, association helps businesses understand customer purchasing habits.

4. Clustering

Clustering is a technique used to group similar data into clusters. It helps in understanding the structure and distribution of data.

This technique:

- Uses visual methods such as graphs and charts
- Shows how data is distributed

- Uses colours or graphics to represent groups

For example:

Grouping customers based on age, income, or buying behaviour.

Clustering helps organizations:

- Identify target customers
- Improve marketing strategies
- Segment large datasets

Thus, clustering is useful in business analytics and decision making.

5. Prediction

Prediction is a powerful data mining technique used to forecast future outcomes.

It uses patterns from past and present data to predict future trends.

This is a part of predictive analytics.

For example:

- Predicting sales
- Weather forecasting
- Stock market prediction
- Customer behaviour prediction

Prediction helps organizations:

- Plan future strategies
- Reduce risks
- Improve decision making

Therefore, prediction plays an important role in business and scientific research.

Data Mining Problems

Data mining systems depend on databases for input data. However, real-world databases are usually large, incomplete, noisy, and dynamic. Because of this, several problems arise during the data mining process.

1. Limited Information

Many databases are designed for specific purposes such as transaction processing and not for data mining.

Therefore:

- Important attributes required for analysis may be missing.
- Some useful information cannot be collected from the real world.
- This makes the learning and analysis process difficult.

Example:

A customer database may not contain behavioural or psychological details, which are useful for marketing analysis.

2. Noise and Missing Values

Real world data is often inaccurate due to:

- Human errors
- Measurement errors
- Data entry mistakes

Some values may also be missing or incomplete.

This leads to:

- Wrong classification
- Poor prediction results
- Reduced accuracy of models

Therefore, data cleaning becomes very important.

3. Uncertainty

Uncertainty refers to:

- Errors in data
- Degree of noise
- Lack of precision

If data is uncertain or imprecise, it affects:

- Pattern discovery
- Accuracy of results

Hence, data precision and reliability are very important in data mining.

4. Size, Updates, and Irrelevant Fields

Databases are:

- Very large in size
- Continuously updated
- Contain irrelevant or unnecessary attributes

Problems caused:

- High storage and processing cost

- Slow performance
- Difficulty in extracting useful patterns

Therefore, data reduction and feature selection techniques are required.

Thus, data mining faces several problems such as limited information, noisy data, uncertainty, and large dynamic databases. These problems must be handled properly to improve the quality of results.

Issues and Challenges in Data Mining

Data mining involves extracting useful knowledge from large and complex datasets. However, real-world data creates several challenges for effective mining.

1. Noisy and Incomplete Data

Real-world data is:

- Heterogeneous
- Incomplete
- Inaccurate

This happens due to:

- Instrument errors
- Human mistakes
- System failures

This reduces the reliability and accuracy of data mining results.

2. Distributed Data

In many organizations, data is stored in:

- Different databases
- Different locations
- Cloud and Internet environments

Problems:

- Difficult to collect all data in one place
- Security and organizational restrictions
- Integration challenges

This makes data mining complex.

3. Complex Data

Modern data includes:

- Images
- Audio
- Video
- Text
- Multimedia

Handling such data is difficult because:

- It requires advanced techniques
- Traditional methods are not suitable

4. Performance

The efficiency of data mining depends on:

- Algorithms
- Hardware
- Data size

If algorithms are not efficient:

- Processing becomes slow
- Results may not be accurate

Thus, performance optimization is very important.

5. Data Visualization

The final results must be presented in a clear and understandable way.

Challenges:

- Difficult to represent complex patterns
- Users may not understand technical results
- Requires effective graphs, charts, and dashboards

Proper visualization helps in better decision making.

6. Data Privacy and Security

Data mining may lead to:

- Privacy violations
- Unauthorized access
- Misuse of sensitive information

Example:

- Medical and financial data

Therefore, strong security and ethical guidelines are required.

Data mining faces many challenges such as noisy data, distributed data, complex data, performance issues, visualization problems, and privacy concerns. Addressing these challenges is essential for successful data mining.

Potential Application Areas of Data Mining

Data mining is widely used in many fields to extract useful information and support decision making. It helps organizations understand patterns, predict future trends, and improve performance.

The major application areas of data mining are:

1. Retail and Marketing

In the retail and marketing sector, data mining helps in understanding customer behaviour and improving sales strategies.

- **Identify buying patterns of customers**
Helps companies know what products customers buy frequently.
- **Find associations among customer demographic characteristics**
Example: Understanding which age group prefers certain products.
- **Predict response to marketing or mailing campaigns**
Helps in targeting customers who are more likely to respond.
- **Market Basket Analysis**
Identifies products that are purchased together.
Example: Customers who buy bread may also buy butter.

Thus, data mining helps retailers improve customer satisfaction and increase profit.

2. Banking

In the banking sector, data mining is mainly used for risk management and fraud detection.

- **Detect fraudulent credit card usage**
Identifies unusual or suspicious transactions.
- **Identify loyal customers**
Helps banks retain valuable customers.
- **Predict customers likely to change credit card affiliation**
Helps in customer retention strategies.
- **Determine credit card spending patterns**
Helps banks offer personalized services.
- **Find hidden correlations among financial indicators**
Helps in financial planning and decision making.
- **Identify stock trading patterns from historical data**
Supports investment and trading strategies.

3. Insurance and Healthcare

Data mining helps insurance and healthcare organizations improve services and reduce risks.

- **Claims analysis**
Identifies which medical procedures are often claimed together.

- **Predict customers likely to buy new insurance policies**
Helps in marketing and customer targeting.
- **Identify risky customers**
Helps reduce financial losses.
- **Detect fraudulent behaviour**
Prevents insurance fraud.

4. Transportation

In transportation and logistics, data mining helps in improving efficiency and reducing costs.

- **Determine distribution schedules**
Helps in planning delivery routes and timing.
- **Analyse loading patterns**
Improves vehicle capacity and resource usage.

Thus, it improves operational efficiency.

5. Medicine

In the medical field, data mining is used for diagnosis, treatment, and patient management.

- **Characterise patient behaviour**
Helps predict patient visits and health conditions.
- **Identify successful treatments**
Helps doctors choose effective therapies for different diseases.
- **Early disease detection**
Improves healthcare quality and saves lives.

Data mining has wide applications in retail, banking, healthcare, insurance, transportation, and medicine. It helps organizations make better decisions, improve efficiency, detect fraud, and provide better services.

Data Warehouse

According to Bill Inmon, the father of Data Warehouse:

A **data warehouse** is defined as a **subject-oriented, integrated, time-variant, and non-volatile collection of data** that supports management in decision making.

In simple words, a data warehouse is a large storage system used to collect and organize data from different sources so that it can be analyzed for better business decisions.

Characteristics of Data Warehouse

1. Subject-Oriented

A data warehouse is organized around major subjects such as:

- Customers
- Products
- Sales
- Suppliers

Instead of focusing on daily operations, it focuses on:

- Analysis
- Decision making
- Reporting

It provides a clear and simple view of specific subjects and removes unnecessary data that is not useful for decision support.

Example: A retail company may create a data warehouse mainly to analyze customer buying behaviour.

2. Integrated Data

A data warehouse collects data from different heterogeneous sources such as:

- Relational databases
- Flat files
- XML files
- Transaction systems
- Network and hierarchical databases

Since these sources have:

- Different formats
- Different coding
- Different naming conventions

The data must be standardized before storage.

This process includes:

- Data transformation
- Data cleaning
- Data consolidation
- Data integration

This reduces:

- Redundancy
- Inconsistency

Thus, the data warehouse provides a consistent and unified view of data.

3. Time-Variant

The data in a data warehouse is stored over a long period of time.

It maintains:

- Historical data
- Current data

Every record contains a time element such as:

- Date
- Month
- Year

This helps organizations to:

- Analyze past trends
- Understand the present situation
- Forecast future outcomes

Example:

Sales data from the last 5 years can be analyzed to predict future sales.

4. Non-Volatile Data

Data in a data warehouse is stable and does not change frequently.

Once the data is loaded:

- It is not updated or deleted
- Only read and analyzed

Unlike operational databases:

- No transaction processing is done
- Data is used only for querying and analysis

Data is updated periodically such as:

- Daily
- Weekly
- Monthly
- Quarterly

Thus, data remains consistent and reliable.

A data warehouse is an important system that stores integrated and historical data. Its subject-oriented, integrated, time-variant, and non-volatile nature helps organizations in decision making, forecasting, and business analysis.

Applications of Data Warehouses

A **Data Warehouse** is widely used in different industries for decision making, forecasting, and strategic planning. It stores large volumes of historical data which helps organizations analyse trends and improve business performance.

1. Consumer Goods Industry

In the consumer goods sector, companies produce and sell products such as food, beverages, electronics, and household items. Data warehouses help them to:

- Analyse **customer buying behaviour** over time.
- Study **seasonal trends** and demand patterns.
- Identify the **most popular products** in different regions.
- Plan production and supply based on past data.
- Improve marketing strategies and promotions.

For example, companies like Hindustan Unilever use data warehouses to understand customer preferences and plan new product launches.

2. Banking Services

Banks generate huge volumes of transaction data daily. A data warehouse helps banks to:

- Analyse customer transactions and spending patterns.
- Detect fraudulent activities and unusual behaviour.
- Evaluate customer creditworthiness.
- Manage risk and regulatory compliance.
- Provide personalized services and offers.

For example, State Bank of India uses data warehousing techniques to analyse customer accounts, loans, and financial transactions.

3. Financial Services

Financial organizations such as stock markets and investment firms use data warehouses for:

- Market trend analysis.
- Risk assessment and portfolio management.
- Monitoring investment performance.
- Predicting future financial conditions.
- Supporting real-time decision making.

For example, companies like Goldman Sachs analyse historical financial data to identify profitable trading strategies.

4. Manufacturing Industry

Manufacturing companies use data warehouses to:

- Monitor production processes and quality.
- Track supply chain and logistics.

- Reduce production costs.
- Identify equipment failures and maintenance needs.
- Improve inventory management.

For instance, Toyota uses data warehouses to improve efficiency and reduce waste.

5. Retail Sector

Retail organizations generate large amounts of sales and customer data. Data warehouses help to:

- Perform **market basket analysis**.
- Understand customer purchasing habits.
- Manage inventory and supply chain.
- Improve customer satisfaction and loyalty.
- Plan marketing campaigns and discounts.

For example, Amazon uses data warehousing to recommend products and improve customer experience.

Advantages of Data Warehousing

A data warehouse provides many benefits to organizations. These advantages improve business performance and decision-making.

1. Cost Efficiency

A data warehouse reduces operational costs by:

- Eliminating duplication of data.
- Improving resource utilization.
- Reducing manual work.
- Supporting automated analysis.

Although initial setup cost is high, long-term savings are significant.

2. High Quality and Reliable Data

Data in a warehouse is:

- Cleaned, integrated, and standardized.
- Free from redundancy and inconsistency.
- Accurate and reliable.

This improves the trustworthiness of reports and decisions.

3. Improved Performance

Data warehouses are designed for fast query processing. This helps:

- Quick generation of reports.
- Faster analysis of large data sets.
- Efficient decision-making.

Unlike operational databases, DW is optimized for analysis, not transactions.

4. Better Decision Making

Managers and executives can:

- Analyse historical trends.
- Compare past and present performance.
- Forecast future outcomes.

This leads to strategic and informed decisions.

5. Consistency and Accurate Data Access

All departments access the **same version of data**, which:

- Reduces confusion and conflicts.
- Ensures consistency across the organization.
- Improves coordination between departments.

6. Time-Saving

Users can quickly access information without searching multiple sources. This:

- Saves time.
- Improves productivity.
- Supports quick business response.

7. Supports Business Intelligence

Data warehouses are the foundation for:

- Data mining
- OLAP
- Business analytics
- Artificial intelligence

These tools help organizations gain competitive advantage.

In modern organizations, data warehousing plays a vital role in **data-driven decision making**. It helps businesses analyse large volumes of data, identify patterns, reduce risks, and improve profitability.

Difference between Data Warehouse and Data Mining

Basis	Data Warehouse	Data Mining
Definition	A data warehouse is a centralized repository that stores large amounts of integrated and historical data for analysis and decision making.	Data mining is the process of extracting useful patterns, trends, and knowledge from large datasets.
Purpose	The main purpose is to store, manage, and organize data in a structured way.	The main purpose is to discover hidden patterns and relationships in data.
Focus Area	It focuses on data storage, integration, and management.	It focuses on knowledge discovery and pattern extraction.
Nature of Work	It is a data preparation and storage process.	It is an analytical and knowledge discovery process.
Sequence	It must be implemented before performing data mining.	It is performed after data is stored in the data warehouse.
Data Sources	Collects data from multiple heterogeneous sources such as databases, files, and online systems.	Uses data stored in warehouses, databases, or other sources.
Data Handling	Data is cleaned, transformed, and loaded periodically.	Data is retrieved, analysed, and interpreted.
Type of Data	Stores historical and time-variant data.	Uses both historical and current data for analysis.
Usage	Used to understand the overall behaviour and trends of business systems.	Used to understand specific patterns and predict future behaviour.
Users	Mainly used by managers, executives, and decision makers.	Used by analysts, data scientists, and researchers.
Performance	Improves system performance by providing organized data.	Improves business output and decision accuracy.
Operations	Data cleaning, extraction, transformation, integration, and loading (ETL).	Classification, clustering, association, regression, prediction, anomaly detection.
Tools Used	OLAP tools, reporting systems, dashboards.	Machine learning and statistical tools.
Example	Companies like Amazon store customer and sales data in warehouses.	The same companies use data mining to recommend products and predict customer behaviour.

Multidimensional Data Model

Data Warehouses (DW) and OLAP tools are based on a **Multidimensional Data Model**. This model represents data in the form of a **Data Cube**, which allows data to be viewed and analysed in multiple dimensions.

Unlike traditional relational databases (which store data in two-dimensional tables), the multidimensional model allows users to analyse data from different perspectives.

Data Cube

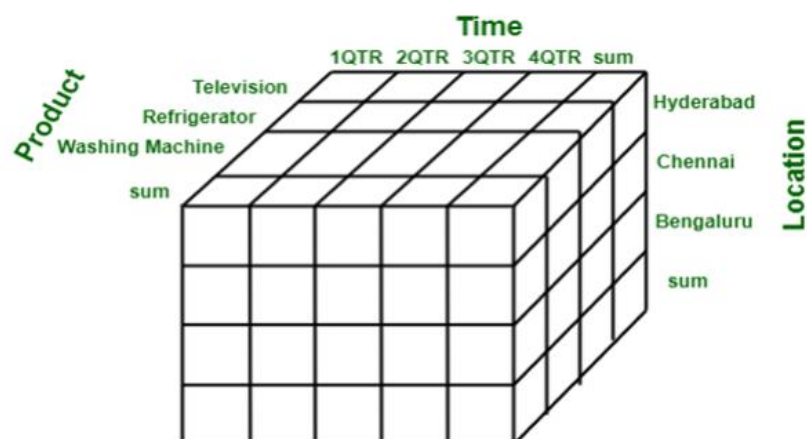
A **Data Cube** is a logical structure that represents data in multiple dimensions.

For example, a sales data cube may have three dimensions:

- **Time**
- **Item**
- **Location**

Each cell in the cube represents a **measure value** such as total sales.

	Location="Chennai"				Location="Kolkata"				Location="Mumbai"				Location="Delhi"			
	item				item				item				item			
Time	Egg	Milk	Bread	Biscuit	Egg	Milk	Bread	Biscuit	Egg	Milk	Bread	Biscuit	Egg	Milk	Bread	Biscuit
Q1	340	360	20	10	435	460	20	15	390	385	20	39	260	508	15	60
Q2	490	490	16	50	389	385	45	35	463	366	25	48	390	256	20	90
Q3	680	583	46	43	684	490	39	48	568	594	36	39	436	396	50	40
Q4	535	694	39	38	335	365	83	35	338	484	48	80	528	483	35	50



This structure allows users to analyse:

- Sales by time
- Sales by product
- Sales by region
- Combination of all dimensions

Components of Multidimensional Model

1. Dimensions

Dimensions are perspectives or entities with respect to which an organization keeps records. They describe the **context** of data.

Example: An electronics company creating a Sales Data Warehouse may use:

- **Time Dimension** (Day, Month, Year)
- **Item Dimension** (Laptop, Mobile, TV)
- **Location Dimension** (City, State, Country)

Each dimension has a **Dimension Table**, which contains descriptive attributes.

Example of Time Dimension Table:

Time_ID	Day	Month	Year
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2. Facts

Facts are numerical measures used to analyse relationships between dimensions. They represent measurable business data.

Example of Sales Facts:

- Dollars_Sold (Total Sales Amount)
- Units_Sold (Number of Units Sold)

These are stored in a **Fact Table**, which contains:

- Fact values (measures)
- Foreign keys of related dimension tables

Example Fact Table:

Time_ID	Item_ID	Location_ID	Dollars_Sold	Units_Sold
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Advantages of Multidimensional Model

- Easy to understand
- Fast query performance
- Supports complex analysis
- Suitable for business intelligence
- Provides multiple perspectives

Definition of OLAP

OLAP (Online Analytical Processing) is a category of software technology that enables analysts, managers, and executives to gain insights into data through fast, interactive, and consistent access to multiple views of information.

OLAP systems:

- Work on Data Warehouse data
- Provide multidimensional analysis
- Support complex queries
- Help in decision making

OLAP Operations / Functions in Multidimensional Model

OLAP supports several operations to analyse data efficiently.

1. Roll-Up (Aggregation)

- Performs aggregation on data cube.
- Moves from detailed data to summarized data.
- Climbs up a concept hierarchy.

Example: Daily sales → Monthly sales → Yearly sales

City sales → State sales → Country sales (It reduces detail level)

2. Drill-Down

- Reverse of Roll-up.
- Moves from summarized data to detailed data.
- Steps down the hierarchy.

Example: Yearly sales → Monthly sales → Daily sales (It increases detail level).

3. Slice

- Selects data on one dimension.
- Produces a sub-cube.

Example: Selecting sales data only for Year = 2025 (Fixes one dimension).

4. Dice

- Selects data on two or more dimensions.
- Creates a smaller sub-cube.

Example: Sales of Laptops in 2025 in Karnataka (Multiple filters applied).

5. Pivot (Rotate)

- Rotates data axes for alternative presentation.
- Changes the view of the cube.

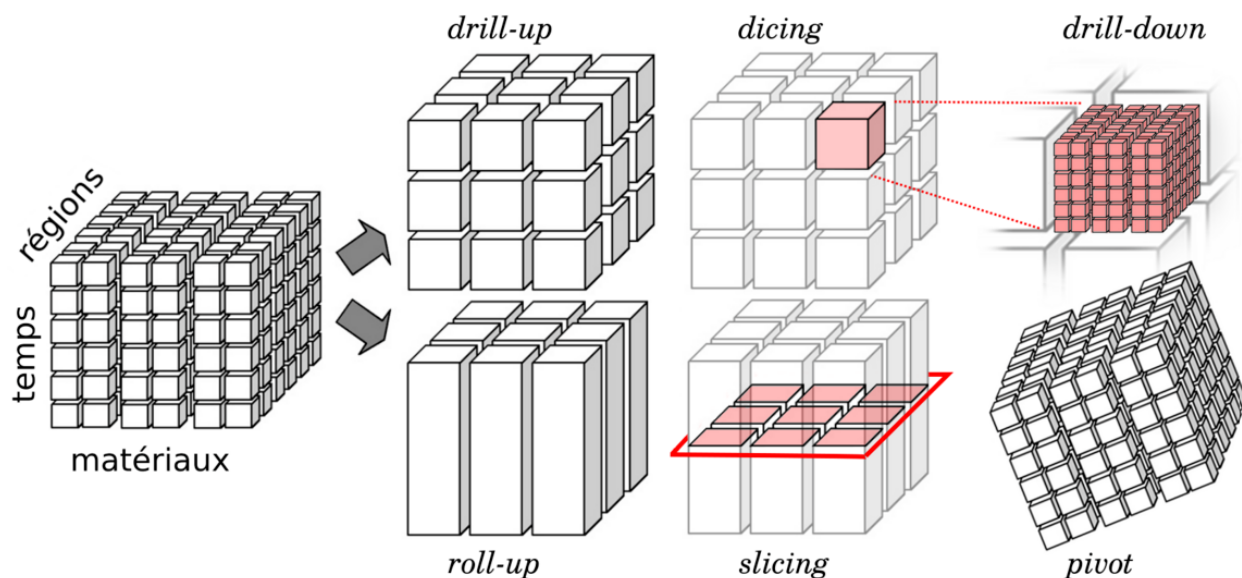
Example:

- Viewing sales by product vs region
- Switching rows and columns in report (Improves data visualization).

6. Drill-Across

- Executes queries involving more than one fact table.
- Combines data from multiple related cubes.

Example: Comparing Sales Data with Inventory Data



Data Preprocessing

Data preprocessing is an important step in **data mining** that involves preparing raw data so that it can be effectively used for analysis and mining. Real world data is often **incomplete, inconsistent, noisy, and redundant**. Data preprocessing improves the **quality, accuracy, and efficiency** of the data before applying data mining algorithms.

The main tasks involved in data preprocessing are:

1. Data Cleaning
2. Data Integration
3. Data Transformation
4. Data Reduction

Data Cleaning

Data cleaning is the process of detecting and correcting errors, noise, missing values, and inconsistencies in data to improve its quality. Hence, cleaning is essential before analysis.

Real-world data is often:

- Incomplete
- Noisy
- Inconsistent
- Redundant

Steps Involved in Data Cleaning

1. Handling Missing Values

Missing data occurs due to errors in data collection or storage.

Methods used -

- Ignore the tuple (record)
- Fill with a constant value (e.g., "Unknown")
- Use mean or median value
- Predict using machine learning models

Example: If salary is missing → replace with average salary.

2. Handling Noisy Data

Noise refers to random errors or incorrect values.

Techniques used -

- Binning
- Clustering
- Regression
- Outlier detection

Example: Incorrect age like 150 years → corrected or removed.

3. Detecting and Removing Outliers

Outliers are abnormal values that differ significantly from others.

Methods:

- Statistical tests
- Distance-based methods
- Visualization techniques

Example: Very high transaction amount may indicate fraud.

4. Resolving Inconsistencies

Inconsistencies arise when data from different sources does not match.

Example: “M” and “Male” or Different date formats

Solution: Standardization and normalization.

5. Duplicate Removal

Remove repeated records to avoid redundancy.

Importance of Data Cleaning

- Improves accuracy
 - Reduces errors
 - Enhances mining results
 - Saves storage and processing time
-

Data Integration

Data integration is the process of combining data from multiple sources into a unified and consistent dataset.

Sources may include:

- Databases
- Flat files
- XML files
- Web data
- Cloud storage

Steps Involved in Data Integration

1. Schema Integration

Combine different database schemas into a single structure.

Example: Customer_ID vs Cust_ID → unified.

2. Entity Identification

Identify same real-world entities from different sources.

Example: Same customer in two systems.

3. Handling Redundancy

Remove duplicate attributes and records.

Example: Address stored in two tables.

4. Data Value Conflict Resolution

Resolve conflicts in values.

Example: Currency differences (₹ vs \$), Different measurement units.

5. Metadata Integration

Combine data definitions and meanings.

Importance of Data Integration

- Provides a unified view
 - Improves decision making
 - Reduces inconsistency
 - Enables large-scale analytics
-

Data Transformation

Data transformation converts data into suitable formats for data mining.
It improves: Accuracy, Efficiency and Compatibility

Steps Involved in Data Transformation

1. Data Normalization

Scale data to a standard range.

Methods used -

- Min-Max normalization
- Z-score normalization

Example: Income range converted to 0–1 scale.

2. Data Aggregation

Combine data to reduce size.

Example: Daily sales → Monthly sales.

3. Generalization

Replace low-level data with higher-level concepts.

Example: City → State → Country.

4. Attribute Construction

Create new attributes from existing ones.

Example: Age from Date of Birth.

5. Smoothing

Remove noise from data.

Methods used -

- Regression
- Clustering.

Importance of Data Transformation

- Improves data quality
 - Makes data suitable for mining
 - Enhances algorithm performance
-

Data Reduction

Data reduction reduces the volume of data while maintaining its analytical value. This makes mining faster, efficient and cost-effective

Steps Involved in Data Reduction

1. Data Cube Aggregation

Summarize data.

Example: Store yearly sales instead of daily.

2. Dimensionality Reduction

Reduce number of attributes.

Methods used -

- Feature selection
- Principal Component Analysis (PCA)

3. Numerosity Reduction

Replace data with smaller models.

Methods used -

- Regression
- Clustering.

4. Sampling

Use a subset of data.

Example: Random sampling.

5. Data Compression

Compress data without losing important information.

Importance of Data Reduction

- Reduces storage
 - Improves speed
 - Simplifies analysis
 - Maintains accuracy
-

Discretization

Discretization is the process of **converting continuous numerical data into discrete intervals or categories**. It reduces the complexity of data and makes it easier to analyze.

Example

Continuous data (Age):

Age
21
25
32
45

After discretization:

Age Range
20–30
30–40
40–50

Thus, numerical values are grouped into **intervals or bins**.

Methods of Discretization

- Binning
 - Histogram analysis
 - Clustering
 - Decision tree analysis
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